

Faheem Ali

Undergraduate Mechanical Engineer

Fali.ug24smme@student.nust.edu.pk | (+92) 3304188776 | [LinkedIn: Link](#) | [Portfolio: Link](#)

Education

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

B.E. in Mechanical Engineering

Expected Graduation: September 2028

CGPA: 3.23/4.00 (4 Semesters)

Relevant Coursework: Fluid Mechanics, Thermodynamics, Mechanics of Materials & Machines, Manufacturing Processes, Heat and Mass Transfer, Controls and Electronics Engineering.

Experience

Smart Agri-Tech Laboratory, SINES(NUST)

Jun 2026 – Present

Engineering Design & Prototyping Intern

Islamabad, Pakistan

- Developed and iterated engineering designs through CAD, rapid prototyping, and Design For Manufacturing (DFM) principles, refining concepts based on testing and performance requirements.
- Gained hands-on experience with 3D printing, laser cutting, engraving, chemical PCB etching, PCB design, and IoT systems integration to transform designs into functional prototypes.

School of Mechanical & Manufacturing Engineering (SMME), NUST

Feb 2026 – May 2026

FYP Technical Associate

Islamabad, Pakistan

- Collaborated with final year students on their FYP focused on enhancing PV solar panel efficiency by integrating a chevron type heat exchanger extracting the excess heat from the PV panel using water as the HX fluid. Contributed to fabricating the complete system, including the chevron heat exchanger, water regulation setup, and solar panel assembly.

Manufacturing Resource Center(MRC), NUST

Jun 2025 – Aug 2025

Production Machining Trainee

Islamabad, Pakistan

- Developed foundational machining skills through hands-on training in precision manufacturing operations on manual lathe and vertical milling machinery.
- Specialized in precision cutting tool manufacturing, transforming raw materials including steel bars and carbide blanks into high-performance cutting tools.

Research

Flow-Field Reconstruction Using ML Models

Sep 2025 – Present

Thesis: Authoring a review paper examining ML models for predicting flow behavior and reconstructing flow fields, analyzing data-driven approaches in computational fluid dynamics.

LNG Regasification Cold Energy Recovery through Organic Rankine Cycle

Feb 2026 – Present

Thesis: Authoring a review paper examining LNG cold energy recovery using Organic Rankine Cycle (ORC) configurations, evaluated from exergy, economic, environmental, and efficiency perspectives.

Projects

PID Controlled Self-Balancing Two-Wheeled Robot

ATMega328 based project utilizing a PID control algorithm, integrating sensor feedback for real-time balance correction.

Car Inventory & Recommendation System

Developed a car recommendation system that matches user requirements against a vehicle database to recommend cars.

Automated Irrigation System

ATMega328 based project integrating soil moisture sensing to automate water delivery with optimized usage.

Mousetrap-Powered Moving car

Built a mechanical car powered by a mousetrap's spring mechanism, converting stored spring energy into motion."

Certifications

Basics of HVAC Systems in Buildings

UniAthena

Complete overview of HVAC systems, their classification based on working mechanisms and first principles of refrigeration and heating cycles.

Research Methodologies

ASME CEME

Beginner guide covering research ethics, persuasive academic writing and technical tools for scientific documentation.

Voluntary Work

Teaching Underprivileged Children

Nov 2025 – Dec 2025

Taught children from Grade 1 to 5 alongside 4 peers as part of a Community Service Course at a local orphanage.

Orphanage Donation

Dec 2025

Contributed to a group donation of PKR 15,000 and distributed shoes to orphaned children at a local orphanage.

Skills

Tools: AutoCAD, Autodesk Inventor, SolidWorks, Additive & Subtractive Manufacturing, Arduino & IoT, Proteus, GitHub.